

Technology Stack

Date	07 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose

1	Frontend / Client-Side	Streamlit	Provides an interactive interface for audio upload, transcription, semantic similarity results, and reports.
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2	Backend / Server-Side	Python	Implements speech processing, semantic analysis, and report generation.
3	Database / Data Storage	No Database (Runtime Data)	Processes uploaded files during runtime without permanent storage.
4	Cloud / Hosting / Deployment	Localhost (Development)	Used for development, testing, and project demonstration.

5	Version Control & CI/CD	Git, GitHub	Supports version control and collaboration.
6	Third-Party APIs / Other Tools	OpenAI Whisper, Sentence-Transformers, Librosa, SoundFile, Matplotlib, ReportLab	Speech-to-text, semantic similarity, audio processing, charts, and PDF report generation.